

Ushtrime ND8ë

1.2.134 Sa humbë nga pesha e vet njeriu i rëndë $Q=800\text{ N}$, nëse pesha e tij specifike është $\gamma_0=1.07 \cdot 10^4\text{ N/m}^3$, ndërsa pesha specifike e ajrit $\gamma_a=12.9\text{ N/m}^3$?

Zgjidhje:

Të dhënat:

$$Q_0 = 800\text{ N}$$

$$\gamma_0 = 1.07 \cdot 10^4 \frac{\text{N}}{\text{m}^3}$$

$$\gamma_a = 12.9 \frac{\text{N}}{\text{m}^3}$$

$$Q = ?$$

$$Q_0 = \gamma_0 V_t \rightarrow V_t = \frac{Q_0}{\gamma_0}$$

$$Q_a = \gamma_a \cdot V_t$$

$$Q_a = \gamma_a \frac{Q_0}{\gamma_0}$$

$$Q_a = 12.9 \frac{\text{N}}{\text{m}^3} \cdot \frac{800\text{ N}}{1.07 \cdot 10^4 \frac{\text{N}}{\text{m}^3}}$$

$$Q_a = 9644.8 \cdot 10^{-4}\text{ N}$$

$$Q_a = 0.96448\text{ N}$$

$$\boxed{Q = \gamma \cdot V}$$

$$Q = mg$$

$$Q = \rho \cdot V \cdot g$$

$$Q = \rho \cdot V$$

$$\boxed{\rho \cdot g = \gamma}$$